Week 6: Sets

Set = "container", order does not matter
defined by what they contain

Defn: A set is a collection of objects.
An object of a set is called an element.
We write this as $a \in S$.

Examples

$$S = \{1, 2, 3, 4, 5\}$$

\{ ... \} in Latex

$1 \in S$, $0 \notin S$, $\pi \notin S$

$T = \{2, 1, 3, 4, 5\} = S$

$T = S$

$\{1, \sqrt{2}\}$, $\{\sqrt{2}, \pi\}$, {David, Jenny, Sarah}

$\{1, 2, \ldots, 10\}$ use "..." to indicate some pattern

\ldots vs ...

$\{2, 4, \ldots, 20\}$

Common Sets

$\mathbb{N} = \{1, 2, 3, \ldots\}$

$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$

$\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ = complex #'s

$\sqrt{2} \notin \mathbb{Q}$, $\sqrt{2} \in \mathbb{R}$

$\sqrt{-2} \notin \mathbb{R}$

$\mathbb{E} = \{\ldots, -4, -2, 0, 2, 4, \ldots\} = 2\mathbb{Z}$

$d \in \mathbb{Z}_{\geq 1}$, we define

$$d\mathbb{Z} = \{\text{'multiples of } d\text{'}\} \quad \boxed{!}$$

More detail

$$= \{\ldots, -2d, -d, 0, d, 2d, 3d, \ldots\}$$

$$= \{dn : n \in \mathbb{Z}\}$$

$$= \{n : n \in \mathbb{Z} \mid d \mid n\}$$

$$= \{n \in \mathbb{Z} \text{ s.t. } d \mid n\}$$

General constructor:

$$\{\text{formula} : \text{parameters} \mid \text{conditions}\}$$

":" = "|" = "such that" = "s.t."

Example: $a, b \in \mathbb{R}$

$$[a,b] = \{x \in \mathbb{R} \text{ s.t. } a \leq x \leq b\}$$

$$[a,b) = \{x \in \mathbb{R} \text{ s.t. } a \leq x < b\}$$

$\boxed{!}$ A set can contain anything

Example:

$$\mathrm{Fun}(\mathbb{R},\mathbb{R}) = \{\text{functions from } \mathbb{R} \to \mathbb{R}\}$$
$$= \{f : \mathbb{R} \to \mathbb{R}\}$$

$g(x) := x^2$, then $g \in \mathrm{Fun}(\mathbb{R},\mathbb{R})$

$h(x) := \sqrt{x}$ $\quad h \notin \mathrm{Fun}(\mathbb{R},\mathbb{R})$

$\quad h \in \mathrm{Fun}(\mathbb{R},\mathbb{C})$

":=" means "definition"

$$\mathbb{Q}[x] := \{a_0 + a_1 x + \cdots + a_n x^n : n \in \mathbb{Z}_{\geq 0} \text{ and each } a_i \in \mathbb{Q}\}$$

$$\mathbb{R}[x] := \{\underline{\qquad\qquad\qquad\qquad\qquad\qquad} \; \mathbb{R}\}$$
$$= \left\{\sum_{i=0}^{n} a_i x^i : n \in \mathbb{Z}_{\geq 0},\ a_i \in \mathbb{R}\right\}$$

$\boxed{!}$ Sets can be elements of sets.

Analogy: Big Amazon box containing many smaller boxes.

Examples:

$$T := \{ \{1\}, \{2\}, \{3,4\} \}$$

T has 3 elements, not 4

$\{1\} \in T$, $\{2\} \in T$, $\{3,4\} \in T$

$\boxed{\{1,2\} \notin T}$

$$S = \{ \{2\}, 3 \}$$

$3 \in S$

$\{2\} \in S$

$2 \notin S$

"$\in$" is not transitive

$x \in y \wedge y \in z \not\Rightarrow x \in z$

$$2 \neq \{2\}$$

$$R = \{ 1, \{1\} \}$$

$1 \in R$

$\{1\} \in R$

Empty set $\longleftrightarrow$ "empty box"

<u>Defn:</u> The empty set $\emptyset$ is the set with the property that $\forall x.\ x \notin \emptyset$. (I.E., $x \in \emptyset$ is always false.)

Defn: We say that 2 sets S and T are equal if $x \in S \iff x \in T$.

(i.e., S & T have the same elements)

Ex: $\{1,2\} = \{2,1\}$

$\{1,2\} \neq \{2,3\}$ b/c $1 \in \{1,2\}$ and $1 \notin \{2,3\}$

$\mathbb{Z} \neq \mathbb{Q}$ b/c $\frac{1}{2} \in \mathbb{Q}$ but $\frac{1}{2} \notin \mathbb{Z}$.

Defn: Let S and T be sets. We say that S is a subset of T if $x \in S \Rightarrow x \in T$. In this case we write $S \subseteq T$. (OR $S \subset T$)

(Equivalently: $\forall x \in S, x \in T$)

To show $S \not\subseteq T$

Find $x \in S$ s.t. $x \notin T$

$\exists x \in S$ s.t. $x \notin T$

Example: $\{1,2\} \subseteq \{1,2,3\}$

$\{1,2,3\} \not\subseteq \{1,2\}$

$3 \in \{1,2,3\}$, $3 \notin \{1,2\}$

Remark: $S = T \iff S \subseteq T$ and $T \subseteq S$

$$\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q} \subsetneq \mathbb{R} \subsetneq \mathbb{C}$$

$-1 \notin \mathbb{N}$, $-1 \in \mathbb{Z}$; $\frac{1}{2} \notin \mathbb{Z}$, $\frac{1}{2} \in \mathbb{Q}$; $\sqrt{2} \notin \mathbb{Q}$, $\sqrt{2} \in \mathbb{R}$; $i \notin \mathbb{R}$, $i \in \mathbb{C}$

Proofs w/ sets

$$P \implies Q$$

Recall "$A \subseteq B$" means $x \in A \implies x \in B$

An implication

Start by "assuming the assumption"

1. "Assume $x \in A$"
2. Write out what "$x \in A$" means

   (I.E. write out the defn)
3. "Argue" or "do calculations"

   $\boxed{?}$
4. Conclude that $x \in B$.

has some defn & in step 3, you verify this

$d\mathbb{Z} = \{n : n \in \mathbb{Z} \mid d \mid n\}$

<u>Prove or disprove</u>:

(i) $6\mathbb{Z} \subseteq 2\mathbb{Z}$

(ii) $2\mathbb{Z} \subseteq 6\mathbb{Z}$

<u>Proof</u>: (ii) This is false b/c $2 \in 2\mathbb{Z}$, but $2 \notin 6\mathbb{Z}$ (b/c $6 \nmid 2$).

(i) Let $x \in 6\mathbb{Z}$. Then $x \in \mathbb{Z}$ and $6 \mid x$.

Since $2 \mid 6$, by transitivity, $2 \mid x$. Thus $x \in 2\mathbb{Z}$. $\blacksquare$

$A = \{4^n - 1 : n \in \mathbb{Z}_{\geq 0}\} = \{0, 3, 15, 63, \dots\}$

$B = 3\mathbb{Z}_{\geq 0}$

$:= \{n \in \mathbb{Z}_{\geq 0} \text{ s.t. } 3 \mid n\}$

We know from week 2 that $3 \mid 4^n - 1$. IE $A \subseteq B$

<u>Claim:</u> $A \subseteq B$.

<u>Proof:</u> Let $x \in A$. Then $\exists n \in \mathbb{Z}_{\geq 0}$ s.t. $x = 4^n - 1$.

By week 2, $3 \mid 4^n - 1$. Thus $4^n - 1 \in 3\mathbb{Z}$. $\blacksquare$

Converse? Is $B \subseteq A$? NO!

$6 \in B$ but $6 \notin A$.

Lemma: Let A, B, C be sets. Suppose that $A \subseteq B$ and $B \subseteq C$. Then $A \subseteq C$.

$x \in A \Rightarrow x \in C$

Proof: Assume $A \subseteq B$ and $B \subseteq C$. Let $x \in A$. Since $A \subseteq B$, $x \in B$. Since $x \in B$, and $B \subseteq C$, $x \in C$. $\blacksquare$

ϕ is the set s.t. "$x \in \phi$" is false $\forall x$.

Claim: $\forall$ set A, $\phi \subseteq A$.

Proof: "There is nothing to check" $\square$

Is every $x \in \phi$ also $x \in A$? Yes...

Contradiction: Suppose $\phi \not\subseteq A$.

I.E. suppose that $\exists x \in \phi$ s.t. $x \notin A$.

Since $x \in \phi$ is always false, we found a contradiction. $\square$

You can't disprove $\phi \subseteq A$.

($\phi \subseteq A$ means $\boxed{x \in \phi \Rightarrow x \in A}$)

Contrapositive: $x \notin A \Rightarrow x \notin \phi$.

Suppose $x \notin A$. Well..... $x \notin \phi$ is true. $\square$

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